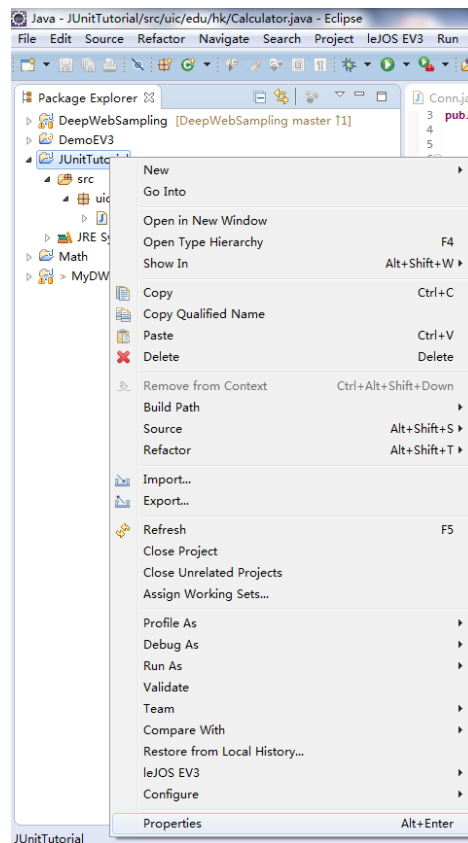


How to do unit test using Junit?

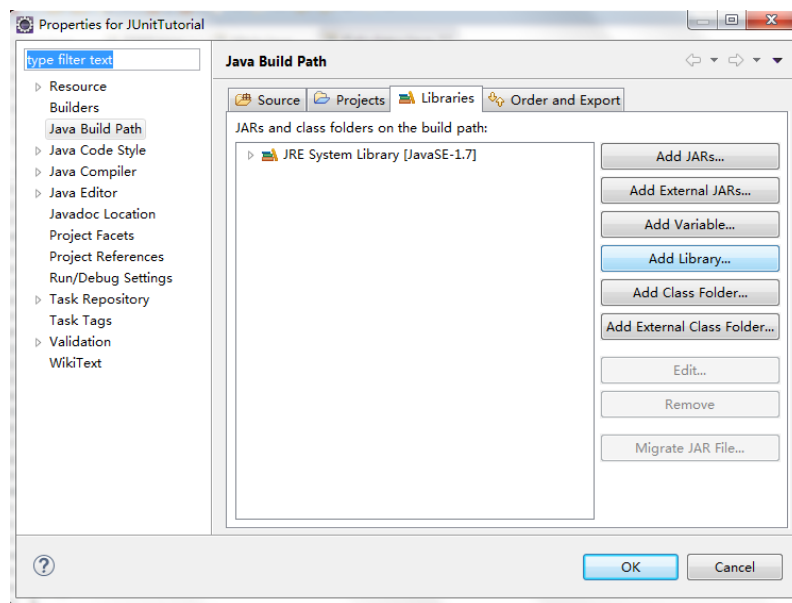
First we create a project called “JUnitTutorial” and write a class for it. I leave few bugs on purpose for demo.

```
public class Calculator {  
    private static int result;  
    public void add(int n) {  
        result = result + n; // correct  
    }  
    public void subtract(int n) {  
        result = result - 1; // Bug: correct result is result = result-n  
    }  
    public void multiply(int n) {  
    } // unfinished method  
    public void divide(int n) {  
        result = result / n;  
    }  
    public void square(int n) {  
        result = n * n;  
    }  
    public void squareRoot(int n) {  
        for (;;)   
            ; // Bug : dead loop  
    }  
    public void clear() {  
        result = 0;  
    }  
    public int getResult() {  
        return result;  
    }  
}
```

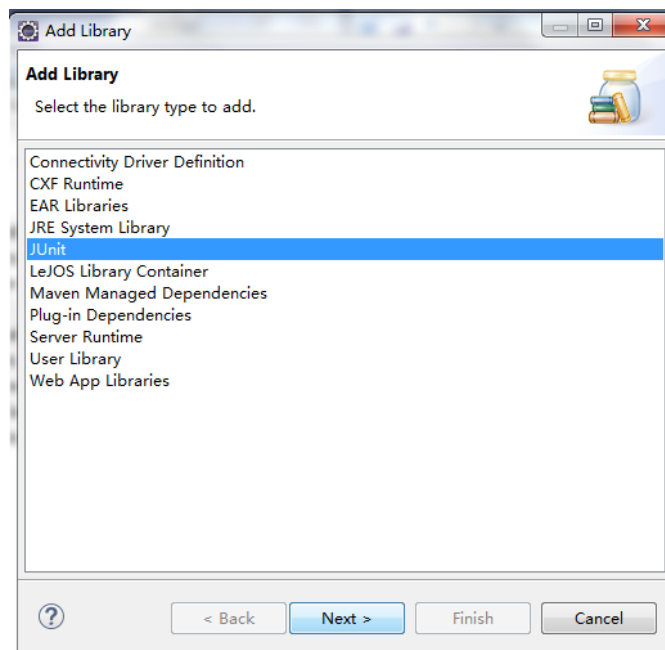
Now we need include Junit library into our project. Right click on the project, click “Properties”.



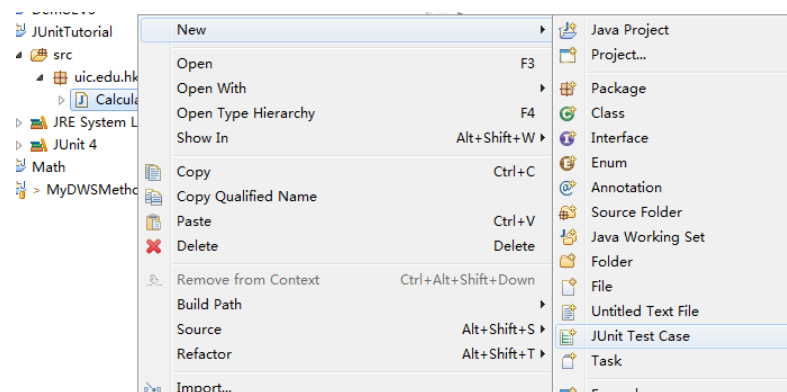
Click “Java Build Path” -> “Libraries”.



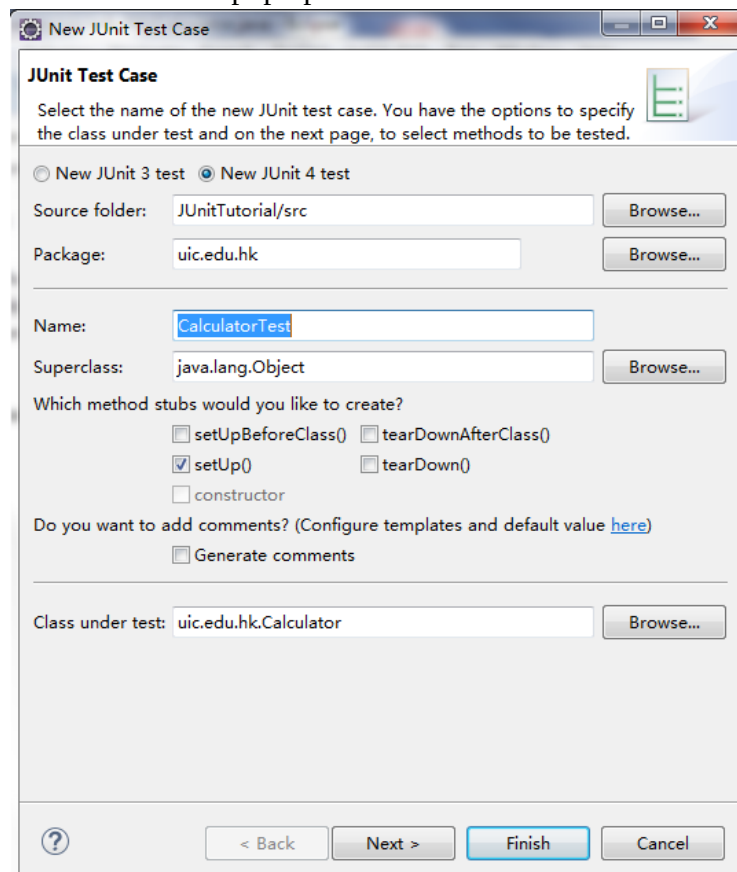
Click “Add Library...” and select JUnit4.



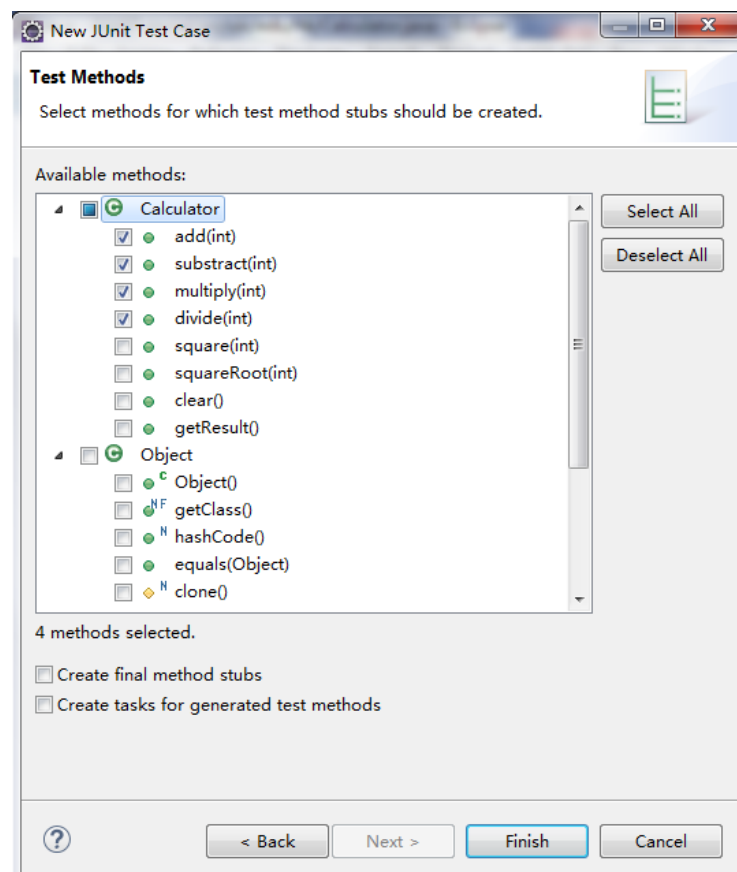
Then, generate Junit testing frame. Right click on our Calculator.java, select “New” -> “JUnit Test Case”.



Then, you can see this window pop up.



Click Next, and select test methods. In this tutorial, we only test the first 4 methods.



Then system will generate a new class called “CalculatorTest.java”. Now we need modify the test codes.

```
import static org.junit.Assert.*;
import org.junit.Before;
import org.junit.Ignore;
import org.junit.Test;

public class CalculatorTest {

    private static Calculator calculator = new Calculator();

    @Before
    public void setUp() throws Exception {
        calculator.clear(); //initialize environment
    }

    @Test
    public void testAdd() {
        calculator.add(2); // add 2
        calculator.add(3); // add 3
        assertEquals(5, calculator.getResult()); //expect answer should be 5
    }

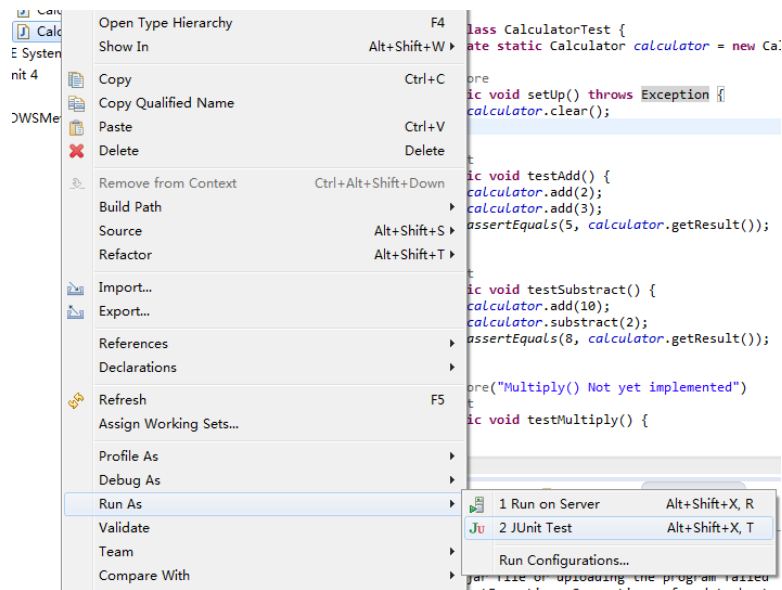
    @Test
    public void testSubstract() {
        calculator.add(10);
        calculator.substract(2);
        assertEquals(8, calculator.getResult()); //expect answer should be 8
    }

    @Ignore("Multiply() Not yet implemented")
    @Test
    public void testMultiply() {
    }

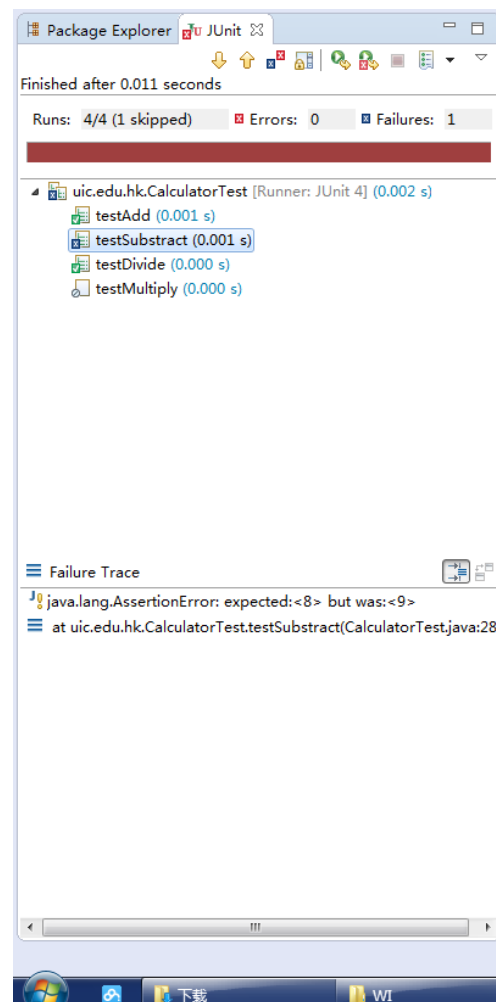
    @Test
    public void testDivide() {
        calculator.add(8);
        calculator.divide(2);
        assertEquals(4, calculator.getResult()); //expect answer should be 4
    }

}
```

After we modified these codes, we can run this class. Right click on “CalculatorTest.java”, select “Run As” -> “Junit Test” to run our test.



Then we can get result.



From the result we can found that, testAdd and testDivide are correct. We ignore testMultiply manually due to we haven't finished it yet. But testSubtract is incorrect so that we can know this method has bug in it. The red progress bar on the top means failures or errors occur.